

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester III)
Scala For Data Science

Practical XXV

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Class: SYBsc IT	Subject: Scala For Data Science
Date of Assignment: 01/08/2026	Date/Time of Submission: 01/09/2026

AIM: Handle missing values in a dataset . Replace the missing values with the column mean.

CODE:

```
build.sbt  IQRA.scala x
1  > object IQRA extends App {
2
3      val sales: List[java.lang.Double] = List(100.0, 200.0, null, 300.0, null)
4      val v = sales.filter(_ != null)
5      val mean = v.map(_.toDouble).sum / v.length
6
7      println("Mean Sales: " + mean)
8
9      val n = sales.map { value =>
10         if (value == null) mean
11         else value
12     }
13
14     println("Original Data:")
15     println(sales.mkString(", "))
16
17     println("After Replacing Missing Values:")
18     println(n.mkString(", "))
19     println("IQRA S021")
20 }
21
```

OUTPUT:

```
"C:\Program Files\Java\jdk1.8.0_181\bin\j
Mean Sales: 200.0
Original Data:
100.0, 200.0, null, 300.0, null
After Replacing Missing Values:
100.0, 200.0, 200.0, 300.0, 200.0
IQRA S021
```

Practical XXVI

AIM: Perform logistic regression using breeze library.
Classify a dataset with binary value.

CODE:

```
build.sbt  IQRA.scala x
2      object IQRA extends App {
8          var weight = 0.0
9          var bias = 0.0
10
11         // Sigmoid function
12         def sigmoid(z: Double): Double = {
13             1.0 / (1.0 + math.exp(-z))
14         }
15         // Train the model
16         for (i <- 1 to 1000) {
17             for (j <- 0 until x.length) {
18                 val prediction = sigmoid(weight * x(j) + bias)
19                 val error = prediction - y(j)
20
21                 weight = weight - 0.01 * error * x(j)
22                 bias = bias - 0.01 * error
23             }
24         }
25
26         // Classify the data
27         println("Predictions:")
28
29         for (i <- 0 until x.length) {
30             val probability = sigmoid(weight * x(i) + bias)
31             val result =
32                 if (probability >= 0.5) 1 else 0
33
34             println("Input = " + x(i) + ", Probability = " + probability + ", Class = " + result)
35         }
36         println("IQRA S021")
37     }
```

OUTPUT:

```
Input = 2.0, Probability = 0.16412826954644827, Class = 0  
Input = 3.0, Probability = 0.4149888799701166, Class = 0  
Input = 4.0, Probability = 0.7193154698480789, Class = 1  
Input = 5.0, Probability = 0.9025176083649726, Class = 1  
IQRA S021
```

```
Process finished with exit code 0
```